Lab 5: REST publish-subscribe system

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Due date: See HuskyCT

Objective

This lab's goal is to show how the REST works in conjunction with a publish-subscribe pattern. We will do this with Python Flask.

Basic requirements

1. All subscribers are stored in the Publisher server.
2. There is one endpoint for adding a subscriber – each subscriber is a name and a URL
3. There is one endpoint for deleting subscribers – supplying only the subscriber's name
4. There is one endpoint for returning a list of all subscribers and their URLs
5. There is one endpoint for updating the published subject and notifying all subscribers
    a. The notifications can be print statements on the backend
    b. The backend should be run in one terminal and another terminal should be used for testing with curl

Preparation

See HTTP slides and example code

First run Python on the example code in one codespace terminal.
The next commands should be run in a different codespace terminal.

curl -X POST -H "Content-Type: application/json" -d '{"name":"Grace","URI":"http://grace.example.com"}' http://localhost:5000/add-subscriber

curl -X POST -H "Content-Type: application/json" -d '{"name":"Ada","URI":"http://ada.example.com"}' http://localhost:5000/add-subscriber

curl -X POST -H "Content-Type: application/json" -d '{"name":"Alan","URI":"http://example.com"}' http://localhost:5000/add-subscriber

curl http://localhost:5000/list-subscribers

Grace
Alan
Ada

curl -X DELETE -H "Content-Type: application/json" -d '{ "name":"Grace" }'
https://localhost:5000/delete-subscriber

curl -X DELETE -H "Content-Type: application/json" -d '{ "name":"Bob" }'
https://localhost:5000/delete-subscriber

You may also use the client from your computer to the git codespace:

Windows> curl.exe -X POST -H "Content-Type: application/json" -d '{ "subject-update": "buy IBM"}' https://**obscure-space-halibut-v6q6x5w9r9w2w5jg-5000.app.github**

Where the red url is generated by the github codespace.

Details

You may start with the code:
https://github.com/wonder-phil/cse2102-2026 in the folder flaskPubSub

You must implement this in a github.com CodeSpace

What to hand in

One zip file containing

1. Brief description of what you did (< 1 page)
2. All code
3. A video of your builds and tests running – you should build basic unit tests
4. A README file that shows how to run your system